# AP CSP Unit 6 Day 7.1 Hexadecimal and RGB Assessment

1. (DOK 1) What base is the hexadecimal number system?

A. 2

B. 8

C. 10

D. 16

2. (DOK 1) Which hexadecimal digit represents decimal 15?

A. D

B. E

C. F

D. 10

3. (DOK 1) In an RGB color code, what do the letters RGB stand for?

A. Red Green Blue

B. Red Gray Black

C. Random Green Blue

D. Red Gold Brown

4. (DOK 1) What hexadecimal color code represents white?

A. 000000

B. FFFFFF

C. FF0000

D. 00FFFF

5. (DOK 2) What is the hexadecimal value of decimal 26?

A. 1A

B. 2A

C. A1

D. 1F

6. (DOK 2) Why is hexadecimal commonly used to represent colors on web pages?

A. It uses fewer symbols than binary and is easier for humans to read

B. It stores more colors than RGB

C. It replaces pixels

D. It eliminates binary data

7. (DOK 2) If the RGB values are 255, 255, 0, what color is represented?

A. Cyan

B. Magenta

C. Yellow

D. Blue

8. (DOK 3) A web designer sees the color code #FF00FF. Based on RGB values, which color should appear and why?

A. Yellow because red and green are maximized

B. Magenta because red and blue are maximized

C. Cyan because green and blue are maximized

D. Black because all values are zero

9. (DOK 3) Why is hexadecimal a useful bridge between binary data and human-readable color values?

A. It converts data into analog signals

B. Each hexadecimal digit represents four binary bits, making long binary values easier to interpret

C. It removes the need for binary

D. It increases the number of pixels

10. (DOK 4) A student must create a webpage color from RGB values of (255, 128, 0). Which process best demonstrates understanding of the lesson?

A. Convert each RGB value to hexadecimal and combine them into a six-digit hex color code

B. Convert RGB values directly into ASCII characters

C. Use only binary values without conversion

D. Replace RGB values with decimal-only color names